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SUPPLEMENTARY MATERIAL

Supplementary material for: Quantum Kernels for EEG: An Invariance Mismatch Masquerading as Advantage

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Abstract

Hyperparameter grids for every pipeline, the per-subject differences behind the main benchmark, the kernel-concentration and register-sweep tables, the cross-session and register-sweep figure, and the cross-subject and finite-shot figure and table. Each item is referred to from the main text.

S1 Hyperparameter grids

Table S1. Hyperparameter grids and fixed settings. Every grid is searched by the same inner cross-validation (4-fold within subject; 3-fold subject-grouped for cross-subject transfer; 4-fold stratified within the training session for cross-session transfer). The grids and settings are read from the code that produced the results. γ_0 is the median-heuristic bandwidth, computed on the training split only.

Pipelines	Searched grid
FB-Fidelity-SVM, FB-Fidelity-ref-SVM, FB-HS-overlap-SVM, FB-HS-overlap-ref-SVM, FB-TS+LR, FB-logeuclid-kernel-SVM, FB-riemann-kernel-SVM, Fidelity-SVM, Fidelity-ref-SVM, HS-overlap-SVM, HS-overlap-ref-SVM, PCA-matched-linear, TS+LR, logeuclid-TS+LR, logeuclid-kernel-SVM, riemann-kernel-SVM	$C \{0.1, 1, 10\}$
FB-TS+RBF-SVM, PCA-matched-RBF, TS+RBF-SVM	$C \{0.1, 1, 10\}$; RBF $\gamma \{\text{scale}, 0.01, 0.1\}$
CNOT-kernel-SVM, IQP-kernel-SVM, IQP-no-entangle	$C \{0.1, 1, 10\}$; angle scale $\{0.5, 1, 2\}$
Bures-RBF-SVM, Bures-RBF-ref-SVM, FB-Bures-RBF-SVM, FB-Bures-RBF-ref-SVM, FB-HS-RBF-SVM, FB-HS-RBF-ref-SVM, FB-QRE-RBF-SVM, FB-QRE-RBF-ref-SVM, HS-RBF-SVM, HS-RBF-ref-SVM, QRE-RBF-SVM, QRE-RBF-ref-SVM	$C \{0.1, 1, 10\}$; bandwidth multiplier $\times \gamma_0 \{0.25, 1, 4\}$
CSP+LDA	CSP filters $\{2, 4, 6\}$
FBCSP+LDA	CSP filters per band $\{2, 4, 6\}$
MDM, logvar+LDA	none (no tuned hyperparameter)
Cross-subject and cross-session quantum kernels	$C \{0.1, 1, 10\}$; bandwidth multiplier $\times \gamma_0 \{0.25, 1, 4\}$ for distance kernels

Fixed in every pipeline: OAS shrinkage covariance; LDA with Ledoit–Wolf shrinkage; tangent space at the Riemannian mean with feature standardisation; circuit kernels and dimension-matched controls on a PCA reduction to 4 components scaled to $[0, \pi]$, 2 circuit layers; SVMs on precomputed kernels projected to the positive semi-definite cone; logistic regression with up to 2000 iterations.

S2 Per-subject differences in the sensor-frame benchmark

S3 Kernel concentration versus register size

S4 Cross-session transfer and the register sweep

S5 Cross-subject transfer and finite-shot estimation

S6 Few-trial calibration

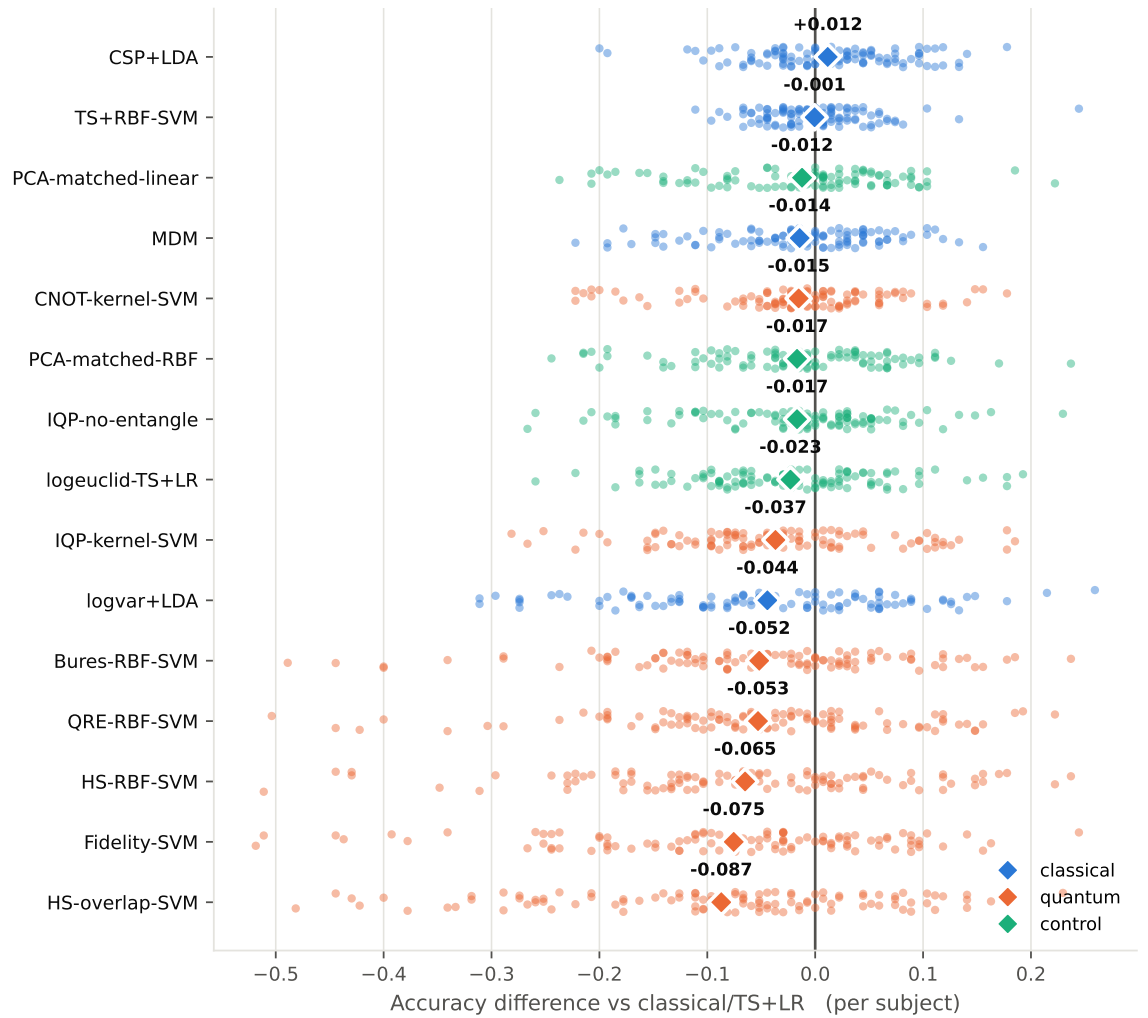


Figure S1. Paired per-subject accuracy differences against the Riemannian tangent-space reference. Dots are individual subjects; diamonds are means. Points left of zero indicate worse performance than the reference.

Table S2. Kernel concentration as a function of register size on real EEG. Register size is swept by varying the channel count over powers of two (4/8/16/32/64 channels = 2–6 qubits), with no change of method. The statistic is the variance of the off-diagonal Gram entries; a factor below one means concentration worsens with scale. The entangled circuit kernel concentrates $2.7\times$ faster in log-slope than the same circuit with entanglers removed.

Kernel	Log-slope / qubit	Factor / qubit	Var (2 qubits)	Var (6 qubits)
IQP-entangled	−0.859	0.424	0.07476	0.00248
IQP-product	−0.319	0.727	0.09396	0.02652
Bures-RBF	−0.195	0.823	0.07758	0.03643
HS-RBF	−0.113	0.894	0.07682	0.05009
Fidelity	+0.493	1.638	0.00030	0.00213
HS-overlap	+0.536	1.709	0.00043	0.00369

Table S3. Off-diagonal Gram variance versus register size in both frames, $n = 104$ subjects, PhysioNet. Larger is better: a kernel whose Gram variance collapses cannot separate trials. “Gain” is reference over sensor at the same register; “vs. 3q” is the reference-frame variance relative to its own three-qubit value. In the reference frame no kernel’s variance falls below its three-qubit value at any larger register, so the concentration that the sensor frame exhibits is not a qubit-count effect once the frame is corrected.

Register	Kernel	Sensor ($\times 10^{-3}$)	Reference ($\times 10^{-3}$)	Gain	vs. 3q
3 q (8 ch)	Bures	0.40	2.14	$5.4\times$	1.00
	Fidelity	0.37	1.76	$4.7\times$	1.00
	HS overlap	0.54	4.92	$9.2\times$	1.00
	QRE	1.86	9.92	$5.3\times$	1.00
4 q (16 ch)	Bures	1.25	5.09	$4.1\times$	2.37
	Fidelity	0.99	3.61	$3.6\times$	2.05
	HS overlap	1.46	9.22	$6.3\times$	1.87
	QRE	6.68	33.02	$4.9\times$	3.33
5 q (32 ch)	Bures	1.10	4.62	$4.2\times$	2.16
	Fidelity	0.94	3.34	$3.6\times$	1.90
	HS overlap	1.30	10.12	$7.8\times$	2.06
	QRE	6.08	30.63	$5.0\times$	3.09
6 q (64 ch)	Bures	2.85	4.99	$1.8\times$	2.33
	Fidelity	2.13	3.52	$1.7\times$	2.00
	HS overlap	3.69	11.77	$3.2\times$	2.39
	QRE	19.81	37.38	$1.9\times$	3.77

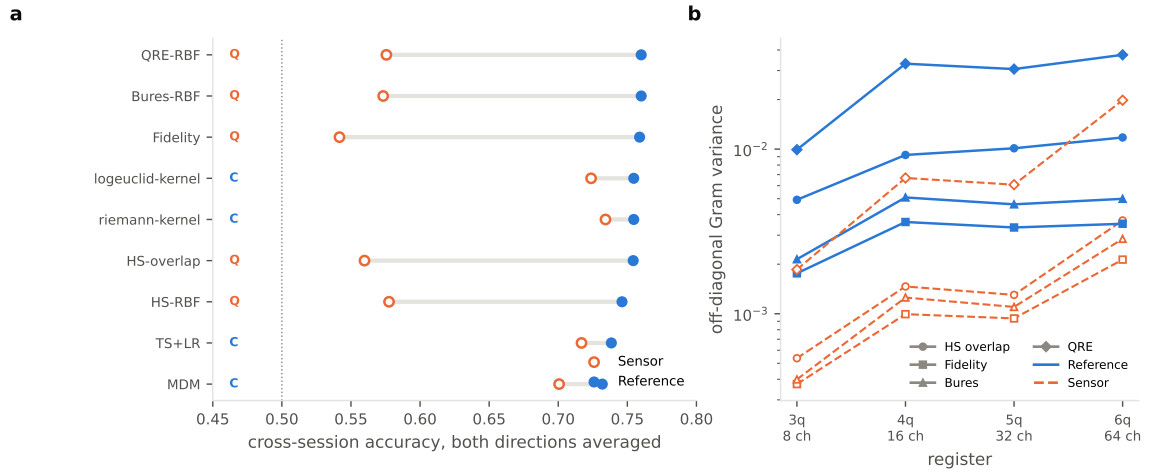


Figure S2. (a) Cross-session transfer within subject on IV-2a, both directions averaged. In the sensor frame the quantum kernels sit near chance while the classical baselines do not; whitening each session by its own label-free mean closes the gap and leaves every SPD-kernel geometry tied. (b) Off-diagonal Gram variance versus register size in both frames on PhysioNet ($n = 104$). Solid, reference frame; dashed, sensor frame. In the reference frame no kernel falls below its three-qubit value at any larger register.

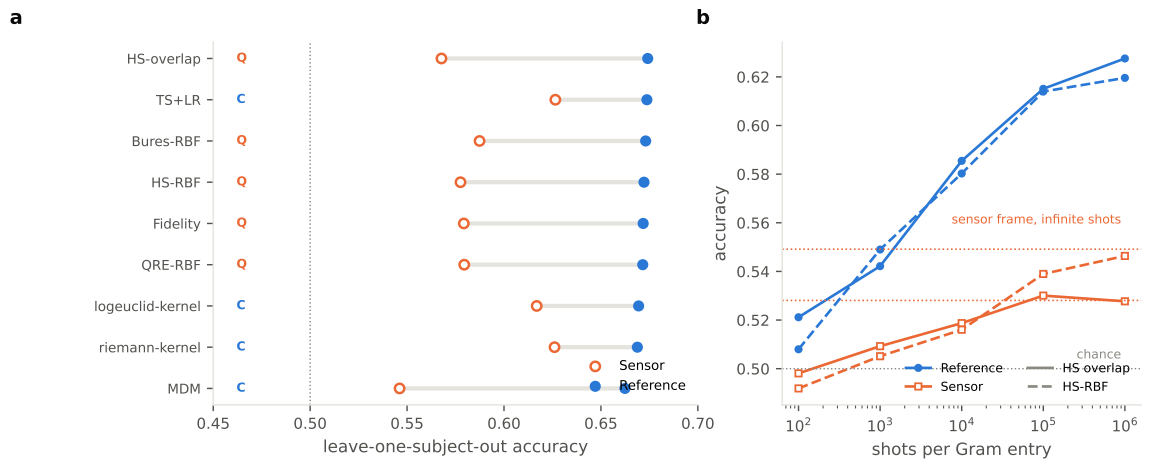


Figure S3. (a) Cross-subject transfer, leave-one-subject-out. Recentring helps every method and helps the quantum kernels (Q) most, but in the reference frame all nine geometries are indistinguishable. (b) Accuracy against shot budget. The reference frame needs more shots to approach its own ceiling because that ceiling is higher, yet from 10^4 shots per Gram entry it already exceeds what the sensor frame attains with unlimited shots.

Table S4. Accuracy under finite-shot estimation. Unlike every other table, the reference state here is estimated from all of a subject's trials, not the training split alone; it uses no labels, but these accuracies are comparable only with one another, not with the nested cross-validation results. $\text{tr}(\rho\sigma)$ is the SWAP-test observable, so S shots give an unbiased estimate with variance $(1 - k^2)/S$; we sample each unordered pair binomially and project the Gram matrix back to the positive semi-definite cone. The reference frame needs *more* shots to approach its own ceiling, because that ceiling is higher, but it dominates the sensor frame in absolute terms from 10^4 shots upwards, above which it exceeds what the sensor frame achieves with unlimited shots.

Kernel	Frame	10^2	10^3	10^4	10^5	10^6	∞
HS-RBF	Sensor	0.492	0.505	0.516	0.539	0.546	0.549
HS-RBF	Reference	0.508	0.549	0.580	0.614	0.620	0.627
HS-overlap	Sensor	0.498	0.509	0.519	0.530	0.528	0.528
HS-overlap	Reference	0.521	0.542	0.585	0.615	0.628	0.631

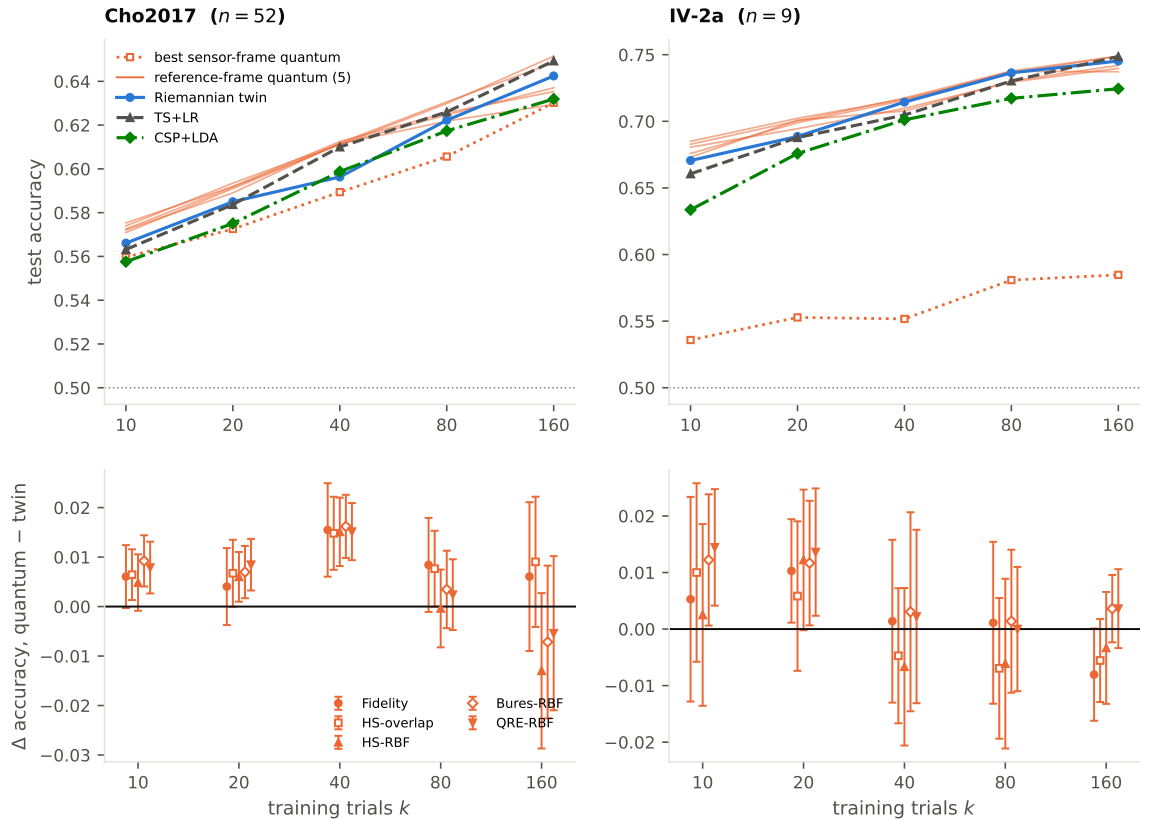


Figure S4. Accuracy against the number of calibration trials, and the quantum minus twin difference that goes with it. The reference-frame kernels hold a small lead over the metric-matched twin while calibration trials are scarce, and it closes as trials are added. The main text gives the sizes, their corrections, and the three limits that bound the result: it lies inside the equivalence margin, it clears the best classical pipeline only at the smallest training set, and the pre-specified trend test does not survive correction.